Prabhat Vilas Karlekar

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Summary

Software Engineer with 3+ years of experience in AI/ML, cryptographic systems, distributed computing, and HPC. IEEE-published researcher in federated learning with homomorphic encryption. Experienced in building production-grade systems spanning distributed systems, network protocol simulation, privacy-preserving ML frameworks, and blockchain infrastructure.

Technical Skills

Languages: Python, C/C++, Rust, Bash/Shell
AI/ML & HPC: TensorFlow, Keras, GRU/LSTM, Scikit-learn, NumPy, Pandas, OpenFHE (CKKS HE), Federated Learning, Apache Kafka, HPC
Blockchain & Cryptography: Smart Contracts, PKI, Merkle Trees, Homomorphic Encryption, Ed25519, SHA
Systems & Networking: Linux, ns-3, Distributed Systems, Computer Networking, CI/CD, GitLab/GitHub, WebAssembly
Backend & Infra: Rust/Axum, Yew/WASM, PostgreSQL, Mongoose HTTP
QA & Testing: STLC, Postman, Pytest, Charles Proxy, UAT, Log Analysis

Experience

Project Engineer – HPC Infrastructure & Ecosystem **Oct 2024 – Present**
Centre for Development of Advanced Computing (CDAC) *Pune, Maharashtra*

- Designed a decentralised architecture integrating blockchain marketplaces with QoS-driven resource allocation for low-latency network services.
- Authored IDFs, whitepapers, and IETF-RFC style protocol specifications for patent-oriented R&D initiatives.
- Architected a privacy-preserving decentralised registry using a Two-Proof Model with Merkle commitments and registry-market separation.
- Built a federated learning framework integrating multikey CKKS homomorphic encryption with TensorFlow/Keras across a 2-client HPC cluster.
- Developed PAYF, an ns-3 based protocol simulation implementing Ed25519-signed shim headers for verifiable per-packet billing.

Quality Analyst – JioCloud **Dec 2023 – Sep 2024**
Reliance Jio Platforms Limited *Navi Mumbai, Maharashtra*

- Designed and executed test scenarios in Agile sprints, reducing regression effort and accelerating deployment cycles.
- Collaborated on defect triage, root-cause analysis, issue reproduction, and release stabilisation across distributed systems.

Graduate Engineer Trainee **Dec 2022 – Nov 2023**
Reliance Jio Platforms Limited *Navi Mumbai, Maharashtra*

- Worked on cloud platforms, DevOps workflows, production monitoring, UAT, and API testing in Agile environments.
- Optimised test cases and contributed to faster issue resolution and improved system stability.

Research & Projects

Privacy-Preserving Quantum-Secure Federated Learning Framework | *OpenFHE, TensorFlow/Keras* **Sep 2025**

- Built an end-to-end federated learning framework integrating Multikey CKKS homomorphic encryption and encrypted model aggregation across HPC nodes.
- Implemented encrypted communication pipelines using Apache Kafka and Mongoose HTTP server for secure weight transfer.

PAYF – Pay As You Forward | *ns-3, Rust, PostgreSQL, Ed25519* **Apr 2026**

- Designed and implemented a cryptographically verifiable packet-forwarding monetisation protocol using ns-3 simulation.
- Built modular networking components including signed QoS shim headers, FlowCache, and settlement engines.
- Integrated PostgreSQL persistence, Rust/Axum APIs, and a Yew/WebAssembly dashboard for settlement visualisation.

GreyMisnomer – Open-Source Verifiable Carbon Credit Registry SDK | *Rust, RFC Design* **2026**

- Designing a Rust-based registry SDK for voluntary carbon markets with protocol-level prevention of double counting and greenwashing.
- Implemented the Two-Proof Model using Proof-of-Integrity (PoI) and Proof-of-Offset (PoO) lifecycle guarantees.

Parameter Optimisation & Feature Engineering for Time-Series ML on HPC **May 2025**

- IEEE-published research presented at SCI 2025 on hyperparameter optimisation and feature engineering for time-series forecasting in HPC environments.

Blockchain via Stealth Addresses: A PKI Approach **Nov 2024**

- Designed a stealth address system using PKI to achieve transaction privacy and unlinkability in decentralised networks.

Blockchain-Based E-Voting System **Jul 2021 – Jul 2022**

- Built a tamper-proof blockchain e-voting platform enabling secure and publicly verifiable remote voting.

Education

Dr. Babasaheb Ambedkar Technological University (DBATU) — **Jul 2018 – Aug 2022**
B.Tech in Computer Science & Engineering – CGPA: 9.06 / 10 — *Lonere, Maharashtra*

Certifications & Training

Microsoft - Coursera: Advanced Data Analysis with Generative AI — May 2026
CDAC Pune: Officer Training – Basic on Artificial Intelligence — Jan 2026
Qualcomm Academy: AI Upskilling – Technical Foundation — Jan 2026
CDAC Hyderabad: Machine Learning for Cyber Security — Oct 2025
Google - Coursera: Agile PM, Git & GitHub, Computer Networking, Crash Course Python — 2023
IBM - Coursera: Blockchain Essentials, Intro to DevOps, Data Analysis with Python — 2022–2023
Coursera: Linux Fundamentals, Python Programming — 2023